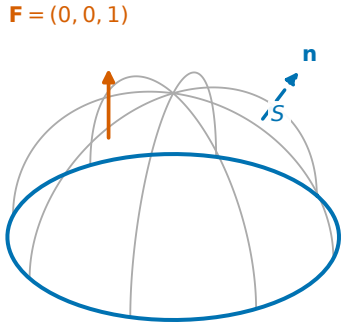
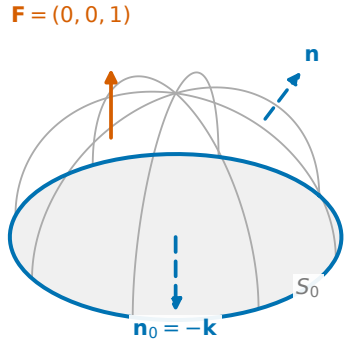
A



$$\Phi_S = \pi$$

B



$$\Phi_{S_0} = -\pi, \quad \Phi_S + \Phi_{S_0} = 0$$